

ENGINEERING TEAM HANDBOOK

Roles · Functions · Career Paths · Assessment Framework

Cordoba Portal · CreditorX · 2025

1. Purpose & Scope

This handbook defines the roles, responsibilities, career levels, and assessment cadences for the in-house software development team at Cordoba / CreditorX. It is designed to help every team member understand what is expected at their current level, what growth looks like, and how the team operates together.

It covers three functional tracks:

- Software Development (Jr → Mid → Sr → Lead)
- DevOps & Infrastructure (Jr → Mid → Sr)
- Quality Assurance / SDET (Jr → Mid → Sr)

It also documents the current team composition and highlights recommended future hires based on the existing Jira workload and growth trajectory.

2. Current Team Composition

The team currently has 7 members covering leadership, backend, mobile, frontend, and support functions.

Name	Role Title	Level	Primary area
Guillermo Malagón	Project Manager / CTO	Lead	Leadership, architecture, delivery
Daniel Rubiano	Senior Developer	Senior	Backend, .NET, integrations
Samil Vargas	Senior Developer	Senior	Mobile (Android / iOS, Kotlin)
Roberto Hiraldo	Senior Developer	Senior	Backend, integrations, Slack/Kazoo
Nicolás Díaz	Junior → Mid Developer	Jr (→ Mid)	Frontend, React, Cordoba Portal
Handersson Pacheco	Junior → Mid Developer	Jr (→ Mid)	Backend, CI/CD, .NET
Diana Adarme	Junior Developer	Junior	QA support, CX triage, documentation

Identified gaps (based on Jira analysis)

- No dedicated DevOps / infrastructure engineer — infra work is currently spread across backend developers
- No Product Manager — roadmap is CTO-driven; a PM would free up technical leadership bandwidth
- QA automation is partially covered but lacks a dedicated SDET role

3. Role Definitions

Each role card below includes a description, key responsibilities, required skills, and KPIs. Roles marked as "open" or "recommended hire" are not currently filled but are recommended based on team workload.

Leadership · Project Manager / CTO [Lead/CTO]  Current team member: Guillermo Malagón	
Overview	Description
Role summary	Leads technical vision, product direction, team health, and delivery. Bridges business goals and engineering execution. Accountable for architecture decisions and team growth.
Key responsibilities	Core skills required
• Define and maintain technical roadmap and architecture • Lead sprint planning, backlog prioritization, and delivery • Hire, mentor, and performance-manage the engineering team • Communicate progress and risks to stakeholders • Set engineering standards, coding conventions, and processes • Evaluate and adopt new tools or technologies • Resolve escalated technical blockers	• System architecture & design patterns • Agile/Scrum leadership • Technical communication • Risk management • People management • Budget & resource planning • Security fundamentals
KPIs / Success metrics	
• On-time sprint delivery rate ≥ 85% • Team retention > 90% annually • Critical production incidents resolved within SLA • Roadmap milestone hit rate • Team satisfaction score (quarterly survey)	

Software Development · Senior Developer — Backend / Full-Stack [Senior]	
 Current team member: Daniel Rubiano	
Overview	Description
Role summary	Designs and implements complex backend systems, leads technical discussions, mentors junior developers, and ensures code quality and architecture coherence across services.
Key responsibilities	Core skills required
- Design and build scalable backend APIs and services - Lead code reviews and enforce engineering standards - Break down epics into well-scoped tasks for the team - Mentor mid and junior developers - Investigate and resolve high-priority production bugs - Write technical documentation and ADRs - Collaborate with PM/CTO on technical feasibility	.NET / C# or Node.js backend - RESTful API design - SQL & NoSQL databases - Redis / caching strategies - CI/CD pipelines - Code review & mentoring - AWS services (EC2, RDS, S3)
KPIs / Success metrics	
- Code review turnaround < 24 hours - Bug recurrence rate on owned modules < 5% - Test coverage on new code ≥ 80% - On-time delivery of assigned sprint tasks ≥ 90% - Junior developer satisfaction with mentoring	

Software Development · Senior Developer — Mobile (Android/iOS) [Senior]	
 Current team member: Samil Vargas	
Overview	Description
Role summary	Owns the mobile application codebase across Android (Kotlin) and iOS. Leads mobile architecture decisions, handles crash triage, and ensures consistent app quality across platforms.
Key responsibilities	Core skills required
Develop and maintain Android (Kotlin) and iOS features	- Kotlin (Android) - Swift / Objective-C (iOS)

- Triage and resolve mobile crash reports and bugs
- Define mobile architecture patterns (OTP flow, auth, offline)
- Own the mobile CI/CD and app store release process
- Coordinate with backend on API contracts
- Review mobile PRs from junior developers
- Keep SDKs, dependencies, and Kotlin/Swift up to date
- Mobile CI/CD (GitHub Actions, Fastlane)
- Push notifications & deep links
- Crash analytics (Firebase, Sentry)
- REST API consumption
- Unit & UI testing (Espresso, XCTest)

KPIs / Success metrics

- App crash rate < 0.1% per release
- App store rating ≥ 4.2
- Mobile bug backlog cleared each sprint $\geq 80\%$
- Zero regressions on OTP / auth flows
- Release cycle on schedule

Software Development · **Senior Developer — Backend / Integrations** [Senior]

 Current team member: Roberto Hiraldo (rhiraldo)

Overview	Description
Role summary	Specializes in backend integrations, Slack-based tooling, third-party APIs, and infrastructure-level bug resolution. Strong focus on reliability, tombstoning logic, and distributed system correctness.
Key responsibilities	Core skills required
• Build and maintain backend integrations (Slack, Kazoo, Forth) • Debug and fix complex distributed system issues • Manage Redis and database-level reliability concerns • Implement and maintain Cordoba Slack app • Review backend PRs and support junior devs • Document integration patterns and runbooks • Monitor and respond to production incidents	• Backend integration patterns • Slack API / bot development • Redis reliability & pub-sub • Webhook handling • Incident response & postmortems • Kazoo / telephony APIs • Database query optimization
KPIs / Success metrics	
• Integration uptime $\geq 99.5\%$ • Incident mean time to resolve (MTTR) < 2 hours • Tombstone/comment accuracy 100% • Zero critical data-loss bugs per quarter	

- Runbook coverage for all owned integrations

Software Development · Mid Developer — Frontend (Portal) [Mid]	
 Current team member: Nicolás Díaz (almost mid → mid)	
Overview	Description
Role summary	Primary owner of the Cordoba Portal frontend (React). Handles all UI sections, i18n, e2e testing, and layout issues. Transitioning from junior to mid with increasing ownership.
Key responsibilities	Core skills required
• Develop and maintain React components for the portal • Implement and fix internationalization (i18n) across all sections • Write and maintain e2e tests for core user flows • Fix frontend bugs and layout/spacing issues • Collaborate with backend on API integration • Participate in code reviews and sprint planning • Document component patterns in Confluence	• React (hooks, context, routing) • TypeScript / JavaScript • i18n / localization libraries • E2E testing (Playwright / Cypress) • CSS / Tailwind • REST API integration • Git workflow & PRs
KPIs / Success metrics	
• UI bug recurrence rate < 10% • E2E test coverage on new features ≥ 70% • i18n coverage 100% for all new strings • Sprint task completion rate ≥ 85% • Peer code review participation weekly	

Software Development · Mid Developer — Backend / Full-Stack [Mid]	
 Current team member: Handersson Felipe Pacheco (almost mid → mid)	
Overview	Description
Role summary	Covers backend features, CI/CD improvements, and error handling across the Cordoba Portal backend. Actively growing into a mid-level role with increasing ownership of infrastructure tasks.
Key responsibilities	Core skills required

- Implement backend features and API endpoints
- Maintain and improve CI/CD pipelines
- Fix backend bugs and improve error handling
- Write and maintain unit and integration tests
- Support DevOps tasks (AWS deployments, Redis)
- Document technical decisions in Confluence
- Participate in code reviews with senior guidance

- .NET / C#
- CI/CD (GitHub Actions)
- AWS basics (EC2, RDS)
- Redis & distributed locking
- SQL / database migrations
- Error handling & logging
- Git & branching strategies

KPIs / Success metrics

- CI/CD pipeline stability $\geq 95\%$
- Backend bug fix turnaround < 48 hours
- Test coverage contribution $\geq 60\%$ on owned modules
- Documentation of new features before release
- Sprint task completion rate $\geq 80\%$

Software Development · **Junior Developer** [Junior]

 Current team member: Diana Adarme

Overview

Description

Role summary

Supports the team with CX triage, testing tasks, documentation, and well-scoped development work. Developing core software development skills under senior mentorship.

Key responsibilities

Core skills required

- Implement well-scoped, clearly-defined tasks
- Triage and document CX support issues
- Write test cases and contribute to QA efforts
- Maintain and update technical documentation
- Participate actively in sprint ceremonies
- Ask questions early, flag blockers quickly
- Support senior developers in debugging sessions

- Basic programming (JavaScript or C#)
- Git fundamentals
- Jira / project management tools
- Documentation (Confluence)
- Bug reproduction and reporting
- Communication & written clarity
- Learning mindset

KPIs / Success metrics

- Assigned tasks completed on time $\geq 75\%$
- Documentation PRs merged each sprint
- Zero unannounced blockers (flag same day)
- Participation in at least 1 code review per sprint
- Skill progression reviewed monthly with manager

DevOps & Infrastructure · DevOps Engineer Jr [Junior]	
 Current team member: (open role — recommended hire)	
Overview	Description
Role summary	Supports the team with basic infrastructure tasks, monitoring setup, and CI/CD maintenance under senior guidance. Entry point into platform engineering.
Key responsibilities	Core skills required
• Maintain and monitor CI/CD pipelines • Support AWS resource provisioning under guidance • Set up basic alerting and dashboards • Write and maintain infrastructure runbooks • Manage Docker containers and images • Help developers with environment setup	• Linux / bash scripting • Docker basics • GitHub Actions or similar CI • AWS fundamentals (EC2, S3, RDS) • Monitoring basics (CloudWatch) • Git • Written documentation
KPIs / Success metrics	
• Runbook coverage for assigned services • Pipeline uptime $\geq 95\%$ • Environment setup time for new devs < 2 hours • Zero unresolved monitoring alerts > 24h	

DevOps & Infrastructure · DevOps Engineer Mid [Mid]	
 Current team member: (future role — 6–12 months)	
Overview	Description
Role summary	Owns CI/CD pipelines, manages AWS infrastructure, implements container orchestration (Kubernetes), and drives observability improvements across services.
Key responsibilities	Core skills required
• Design and manage CI/CD pipelines (GitHub Actions, ArgoCD) • Manage AWS infrastructure (ECS/EKS, RDS, ElastiCache) • Implement and maintain Kubernetes deployments • Build monitoring, alerting, and log aggregation (Datadog/Grafana)	• Kubernetes & Helm • Terraform / IaC • AWS (EKS, VPC, IAM, RDS, ElastiCache) • Docker & container security • Observability (Grafana, Datadog, CloudWatch)

- Manage secrets and environment variables securely
- Conduct infrastructure cost reviews
- Respond to on-call incidents

- Incident management
- Secrets management (AWS Secrets Manager)

KPIs / Success metrics

- Infrastructure availability $\geq 99.9\%$
- Deployment frequency $\geq 2x$ per week
- MTTR for infra incidents < 1 hour
- Cost per environment reviewed quarterly
- All secrets rotated per security policy

DevOps & Infrastructure · DevOps Engineer Sr / Platform Lead [Senior]

 Current team member: (strategic hire)

Overview

Description

Role summary

Leads the platform engineering function, defines infrastructure-as-code strategy, ensures security and compliance, and enables developer productivity at scale.

Key responsibilities

Core skills required

- Define and own cloud architecture strategy
- Lead infrastructure-as-code (Terraform) implementation
- Drive security posture: compliance, pen-test remediation, WAF
- Build internal developer platforms (IDP) and self-service tools
- Capacity planning and cost optimization
- Mentor DevOps Jr/Mid team members
- Lead post-mortems and reliability improvements

- Multi-cloud or AWS advanced (Organizations, Control Tower)
- Terraform / Pulumi
- Security & compliance (SOC2, ISO 27001 basics)
- SRE practices (SLOs, SLAs, error budgets)
- GitOps & deployment strategies (blue/green, canary)
- Networking (VPC, Load Balancers, DNS)
- Cost optimization

KPIs / Success metrics

- SLO achievement $\geq 99.5\%$
- Security vulnerabilities remediated per SLA 100%
- Infrastructure cost within budget ($< 5\%$ variance)
- Developer environment provisioning < 30 min via IDP
- Zero compliance findings in audits

Quality Assurance · QA Engineer Jr [Junior]

 Current team member: (can be current Diana Adarme path)

Overview	Description
Role summary	Executes manual test cases, documents bugs clearly, and learns to write basic automated tests. Ensures features work as expected before release.
Key responsibilities	Core skills required
- Execute manual test plans for new features and bug fixes - Write clear, reproducible bug reports in Jira - Maintain regression test checklists - Support developers in reproducing reported bugs - Document test results in Confluence - Participate in sprint reviews and demos	- Manual testing techniques - Bug reporting (Jira) - Test case writing - Basic automation (Playwright or Selenium basics) - API testing (Postman) - Attention to detail - Clear written communication
KPIs / Success metrics	
- Bug escape rate to production < 5% - Test case documentation per feature 100% - Regression coverage maintained ≥ 80% - Bug report quality score (rated by devs) ≥ 4/5	

Quality Assurance · **QA Engineer Mid / SDET** [Mid]

 Current team member: (recommended hire — aligns with testing label in Jira)

Overview	Description
Role summary	Leads automated test suite development, owns the e2e test strategy, and integrates testing into the CI/CD pipeline. Bridges QA and development through code.
Key responsibilities	Core skills required
- Design and maintain e2e test suites (Playwright/Cypress) - Integrate automated tests into CI/CD pipelines - Define and track QA metrics and coverage targets - Review PRs for testability - Build API test automation (Postman collections / k6) - Conduct exploratory testing on critical flows - Mentor QA Jr on automation best practices	- Playwright / Cypress - API testing & load testing (k6, Postman) - TypeScript/JavaScript for test code - CI/CD integration (GitHub Actions) - Test strategy & planning - BDD (Gherkin) - Performance testing basics

KPIs / Success metrics

- Automated test coverage $\geq 70\%$ on critical paths
- CI pipeline test run time < 10 min
- Bug escape rate to production $< 2\%$
- Flaky test rate $< 3\%$
- Release confidence score (team survey) $\geq 4/5$

4. Career Progression Paths

The following tables define the leveling framework for each functional track. Levels are not purely time-based — promotion criteria must be met consistently over at least one quarter before a level change is considered.

4.1 Software Development Track

Level	Exp. (approx.)	Core competencies	Promotion criteria
Jr Developer	0–2 years	Scoped tasks, basic debugging, documentation	All assigned tasks delivered; clear communication
Mid Developer	2–5 years	Independent modules, code review participation	Owns a domain; mentors juniors; 90% delivery rate
Sr Developer	5–8 years	Architecture input, team mentoring, tech leadership	Leads epics; improves team process; zero regressions
Lead / Architect	8+ years	System design, cross-team coordination, strategy	Drives quarterly roadmap; reduces tech debt measurably

4.2 DevOps & Infrastructure Track

Level	Exp. (approx.)	Core competencies	Promotion criteria
Jr DevOps	0–2 years	CI/CD maintenance, runbooks, Docker basics	All pipelines stable; runbooks written
Mid DevOps	2–5 years	Kubernetes, Terraform, observability, on-call	Owns infra domain; MTTR < 1 h; IaC coverage $\geq 80\%$
Sr DevOps	5–8 years	Platform strategy, security, cost optimization	SLO $\geq 99.5\%$; compliance pass; IDP delivered

4.3 Quality Assurance Track

Level	Exp. (approx.)	Core competencies	Promotion criteria
Jr QA	0–2 years	Manual testing, bug reporting, test cases	Bug escape rate < 5%; full test docs
Mid QA	2–5 years	E2E automation, CI integration, SDET skills	Coverage ≥ 70%; pipeline integrated; flaky < 3%
Sr QA	5–8 years	QA strategy, performance testing, leadership	Org-wide quality metrics owned; team mentored

5. Assessment & Growth Framework

Regular assessments are essential for identifying growth opportunities, resolving blockers early, and keeping the team aligned. The table below defines the recommended assessment cadence.

Frequency	Type	Who	Focus
Weekly	1-on-1 check-ins	All team members	Blockers, morale, progress
Bi-weekly	Sprint retrospective	Full dev team	Velocity, process improvements
Monthly	Individual growth review	Manager + dev	Goals, skill progression, career
Quarterly	Formal performance review	Manager + dev	OKRs, compensation, leveling
Bi-annually	Skills gap analysis	CTO + full team	Team capabilities vs. roadmap
Annually	Culture & org review	All team	Structure, culture, hiring plan

5.1 Assessment principles

- Assessments must result in documented action items — feedback with no follow-up erodes trust.
- Growth conversations are separate from compensation conversations to avoid conflating them.
- Senior developers are assessed on both individual output and team impact (mentoring, process improvements).
- Junior developers should have a written development plan updated monthly.

- Blocked items in Jira that persist > 2 sprints should trigger an assessment conversation.

5.2 Promotion criteria

A developer is ready for promotion when they consistently demonstrate the competencies of the next level for at least one full quarter. Promotion requires:

- CTO/manager endorsement based on performance review
- Peer feedback from at least 2 team members
- KPIs at target level for ≥ 3 consecutive sprints
- A documented growth plan showing readiness for next-level responsibilities

6. Engineering Standards & Processes

6.1 Code quality

- All code must pass peer review before merging to main/develop branches
- Test coverage targets: $\geq 80\%$ for new backend code, $\geq 70\%$ for frontend modules
- No PR merges with failing CI pipeline checks
- Technical debt items must be logged as Jira tasks and triaged each sprint

6.2 Documentation

- All epics must have a Confluence page before development starts
- Architecture decision records (ADRs) required for decisions affecting the system boundary
- API changes must be documented in the same PR that introduces them
- Runbooks required for all production services before deployment to prod

6.3 Security

- No secrets, credentials, or API keys in source code or PRs — use AWS Secrets Manager
- Dependency updates reviewed monthly; critical CVEs patched within 48 hours
- Access control reviewed quarterly by CTO

7. Tools & Technology Stack

Area	Tools / Stack
------	---------------

Backend	.NET (C#), Node.js, Redis, MySQL / AWS RDS, Forth API
Frontend	React, TypeScript, Tailwind CSS
Mobile	Kotlin (Android), Swift (iOS)
Infrastructure	AWS (EC2, RDS, ECS, S3, ElastiCache), Docker, Kubernetes (planned)
CI/CD	GitHub Actions, ECR
QA / Testing	Playwright / Cypress (e2e), Postman (API), unit testing per stack
Monitoring	CloudWatch, Sentry (mobile crash), Datadog (planned)
Project mgmt	Jira (sprints, epics, bugs), Confluence (documentation), Slack

8. Recommended Hiring Roadmap

Based on the Jira backlog analysis (817 issues, 7 current members, identified infra and QA gaps):

Timeline	Role	Priority	Rationale
Now	Jr DevOps Engineer	High	CI/CD and infra work is overloading backend devs
0–6 months	Mid QA / SDET	High	41 bugs in backlog; no dedicated QA automation
6–12 months	Product Manager	Medium	CTO freed from backlog prioritization duties
12–18 months	Mid DevOps (Kubernetes)	Medium	AWS architecture epics planned; need platform owner
18–24 months	Sr Frontend / Second Mid Dev	Low	Portal grows; Nicolás will need support

— End of Handbook —